Phase LEAD & Phase LAG Controller Design

EE352 Automatic Control

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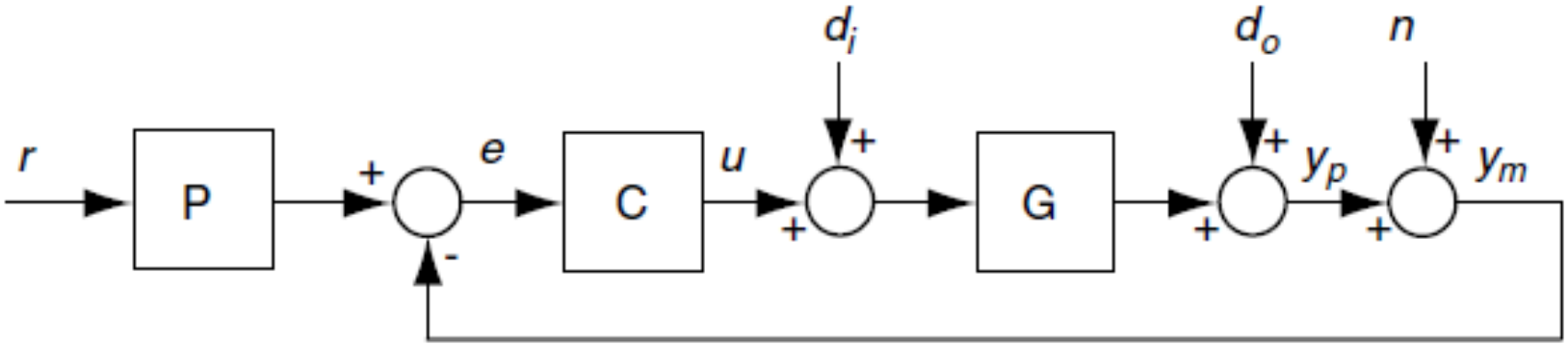
University of Peradeniya
Deep Vision, Bright Future

Design considerations and constraints



With $kL(s) = C(s)G(s)$

$$S(s) = \frac{1}{1 + kL(s)}$$ Sensitivity function

$$T(s) = \frac{kL(s)}{1 + kL(s)}$$ Complementary sensitivity function

$$Y_P(s) = T(s)P(s)R(s) - T(s)N(s) + G(s)S(s)D_i(s) + S(s)D_0(s)$$

Mutually contradictory requirements

- Set point tracking: $T(s)P(s) \approx 1$
- Noise suppression: $T(s) \approx 0$
- Disturbance rejection: $S(s) \approx 0$
- In particular we are constrained by the identity

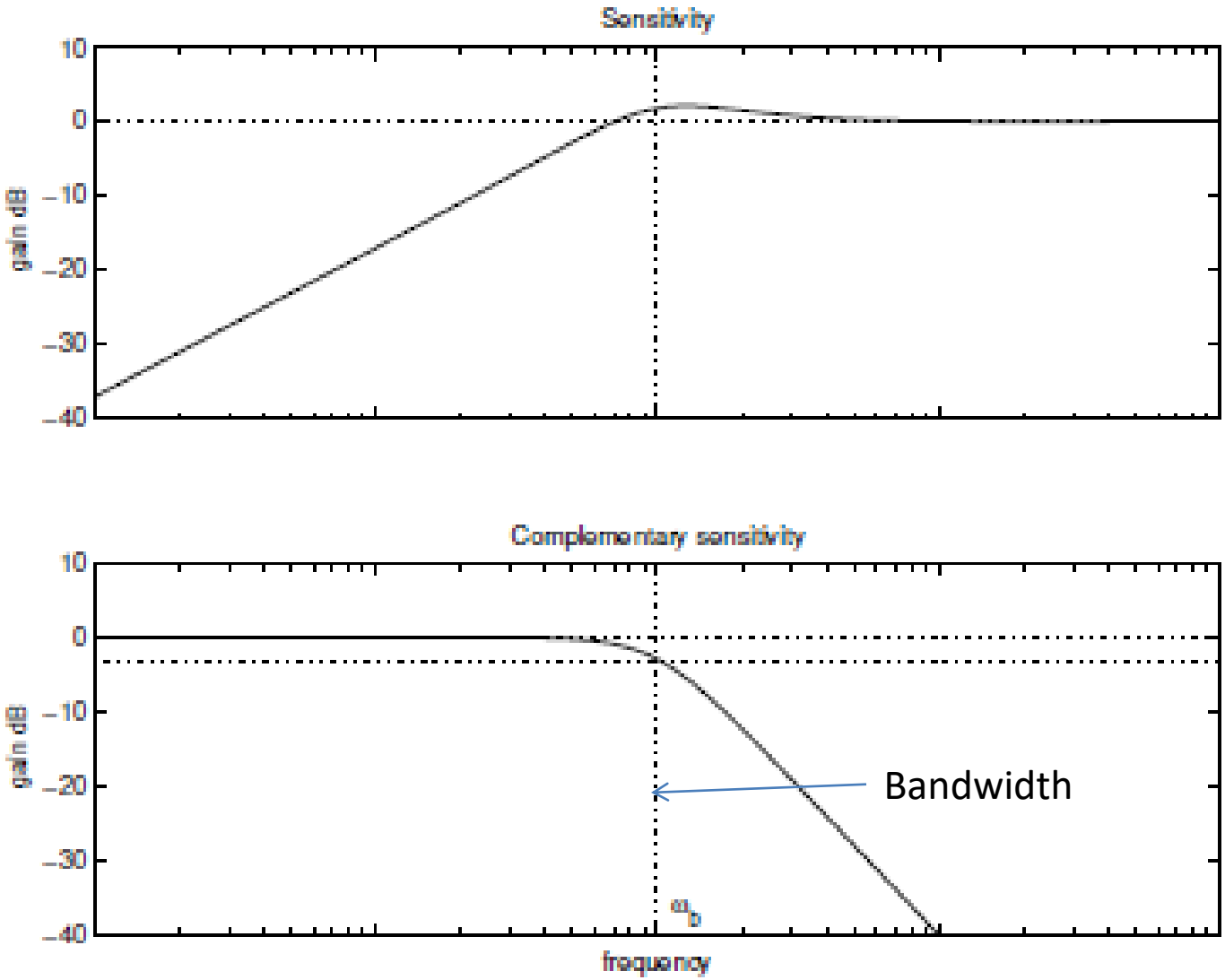
$$S(j\omega) + T(j\omega) = 1, \forall \omega$$

- The usual design compromise is to choose

$S(j\omega) \approx 0 \quad T(j\omega) \approx 1$ at low frequencies

$S(j\omega) \approx 1 \quad T(j\omega) \approx 0$ at high frequencies

Well-designed closed-loop Sensitivity S(jω) and Complementary sensitivity T(jω)



Achieving good closed-loop response

- Choosing $|kL(j\omega)|$ to be large at low frequencies and small at high frequencies
- Care must be taken at the cross-over frequency ω_c. It is usually near to, but a bit smaller than, the closed-loop bandwidth ω_b.
- Varying *k* allows us to choose any cross-over frequency we like.
- However, varying *k* has no effect on the phase, so the phase margin at our chosen *k* may be arbitrarily small or even negative (leading to an unstable closed-loop system).

Phase Margin

- Ensure the phase margin is sufficiently large.
- Rule of thumb: The closed-loop response may often be approximated as a second order transfer function with damping ratio

$$\zeta \approx \frac{\phi_m}{100}$$

- Closed-loop damping ratio of 0.5 or above, we require a phase margin of 50° or above.
- Unfortunately, we cannot increase the phase independent of gain. Nor can we change the shape of the gain independent of phase.

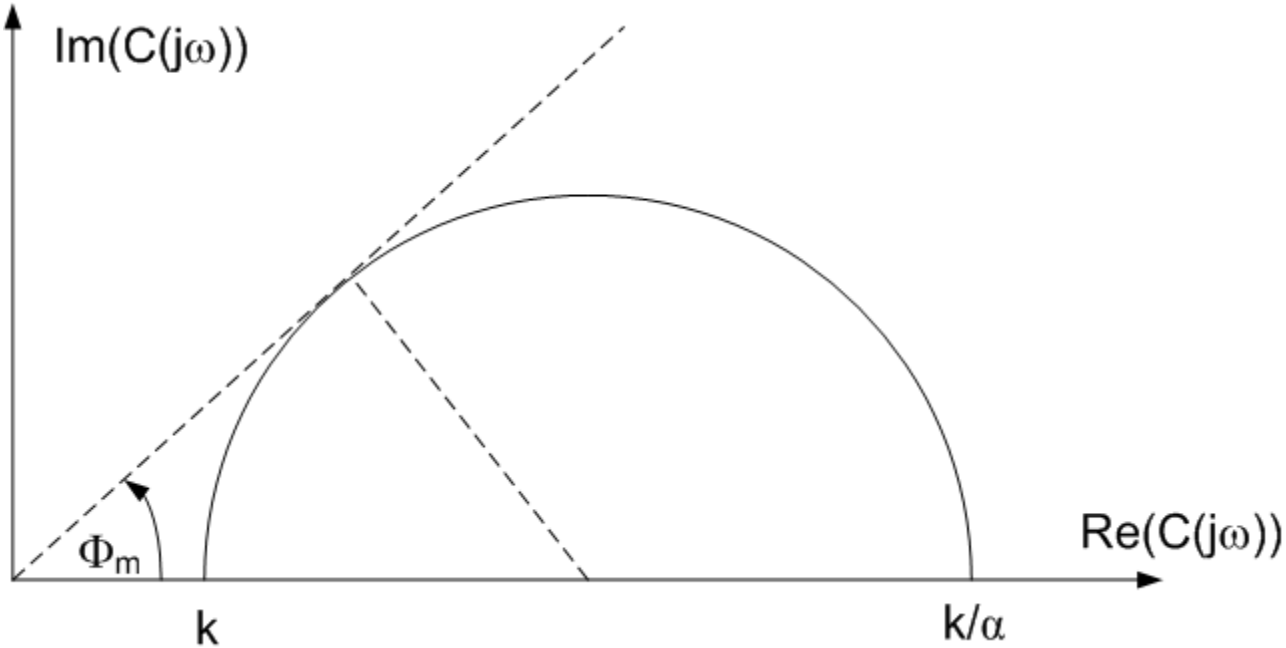
Phase lead design

- A phase lead controller "buys" extra phase margin at the "expense" of increased gain.
- Popular because it can be implemented using passive electronic components.
- It takes the form

$$C(s) = k\frac{1+T_L s}{1+\alpha T_L s}$$

With $\alpha < 1$

Nyquist plot of a Phase Lead controller.



Semicircle with centre at $c = \frac{k}{2\alpha}(1+\alpha)$

Radius $r = \frac{k}{2\alpha}(1-\alpha)$

Maximum phase advance occurs when $\sin\phi_m = \frac{r}{c} = \frac{1-\alpha}{1+\alpha}$

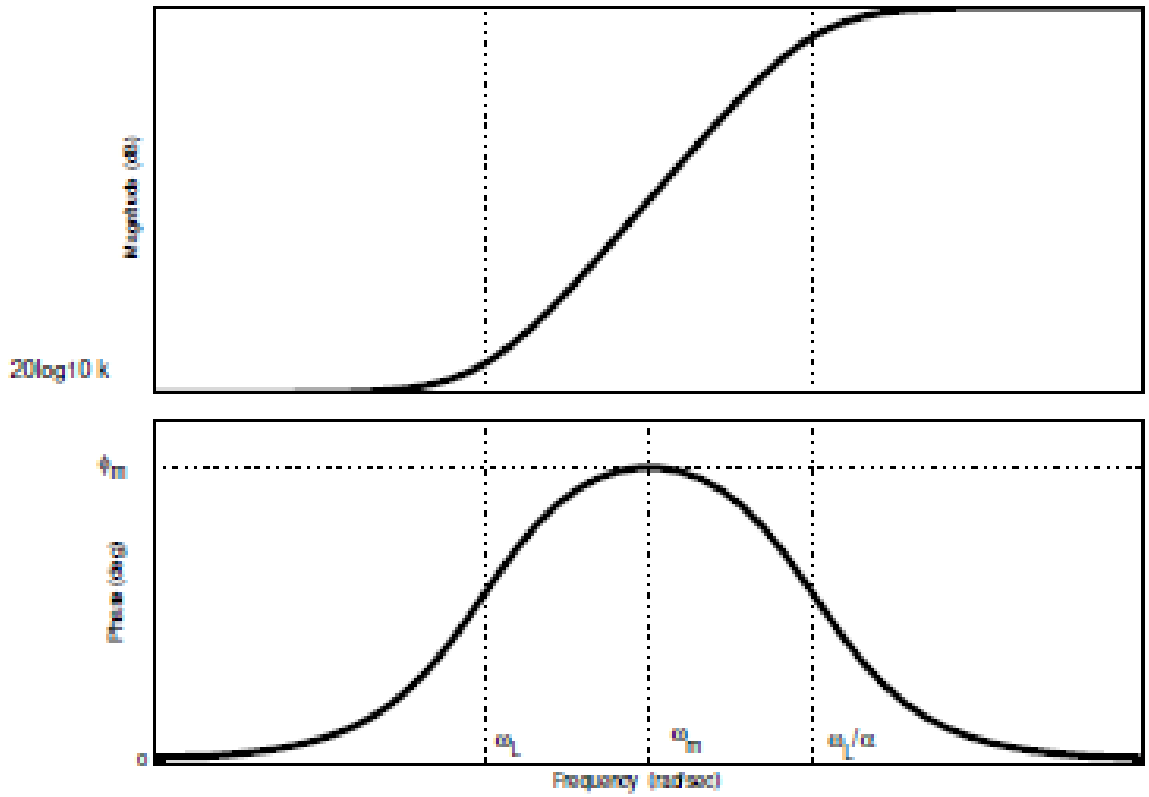
Bode plot of a Phase Lead controller.

$$C(s) = k\frac{1+T_L s}{1+\alpha T_L s}$$

$$\alpha = \frac{1-\sin\phi_m}{1+\sin\phi_m}$$

$$\omega_L = \frac{1}{T_L}$$

$$\sqrt{(\omega_L)\left(\frac{\omega_L}{\alpha}\right)} = \omega_m$$

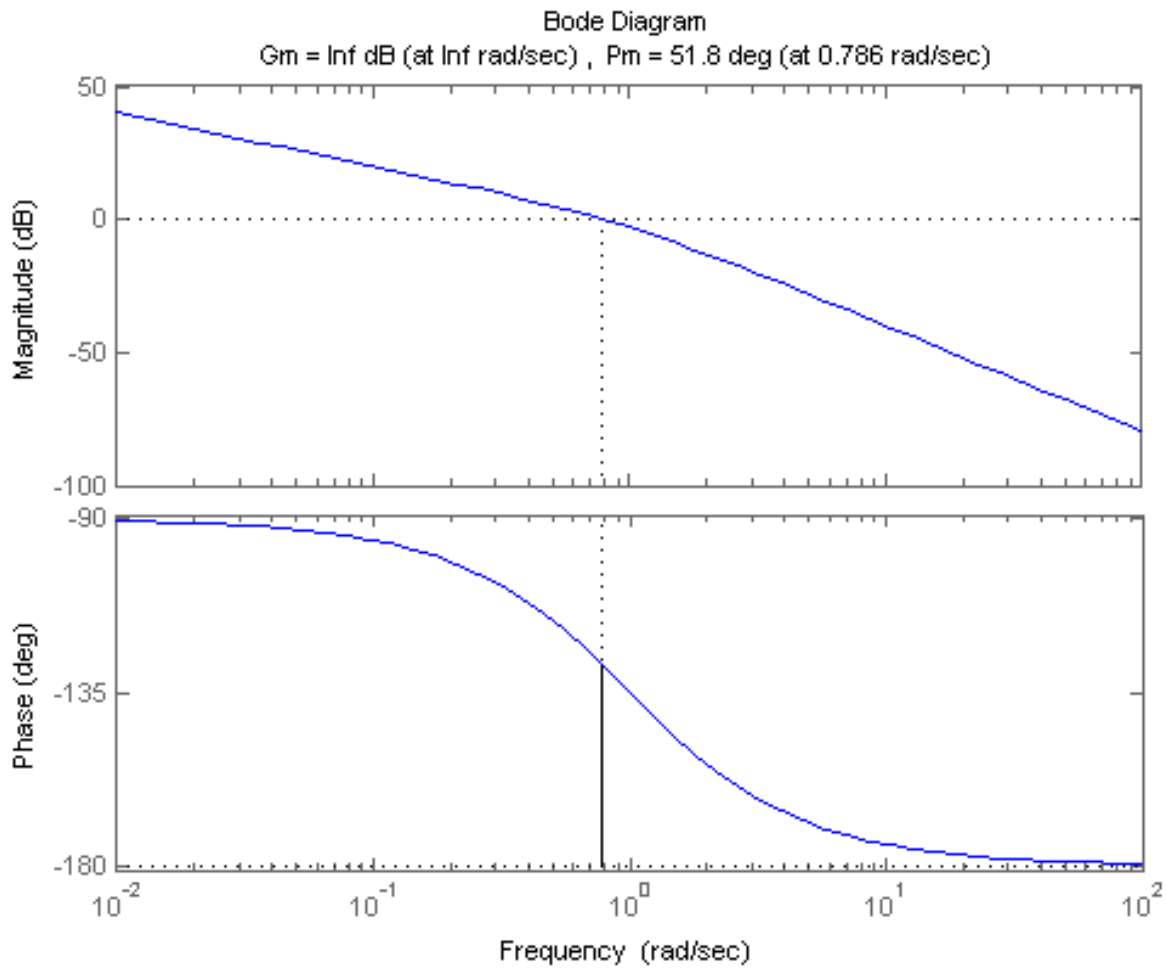


Maximum phase advance occurs when $\omega = \omega_m$

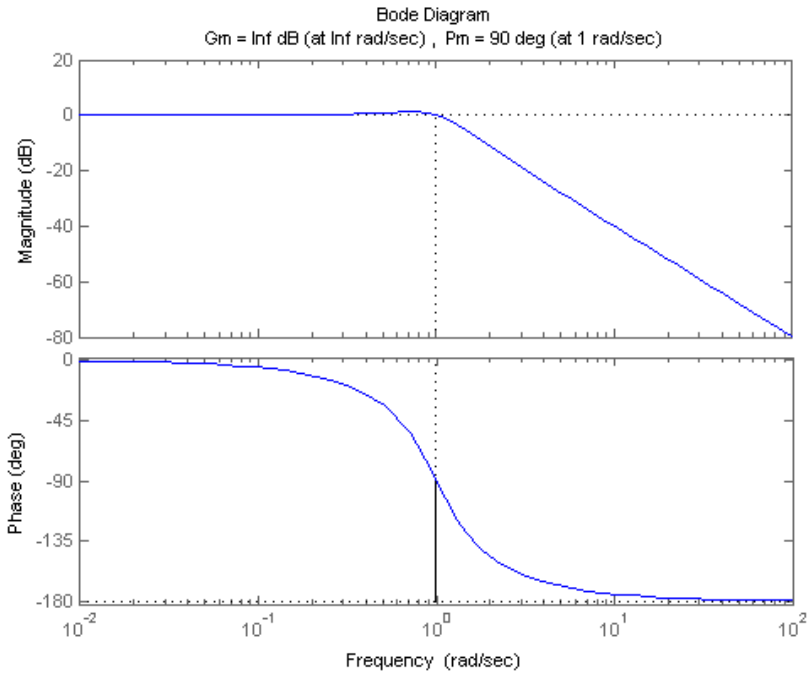
Example

$$G(s) = \frac{1}{s(s+1)}$$

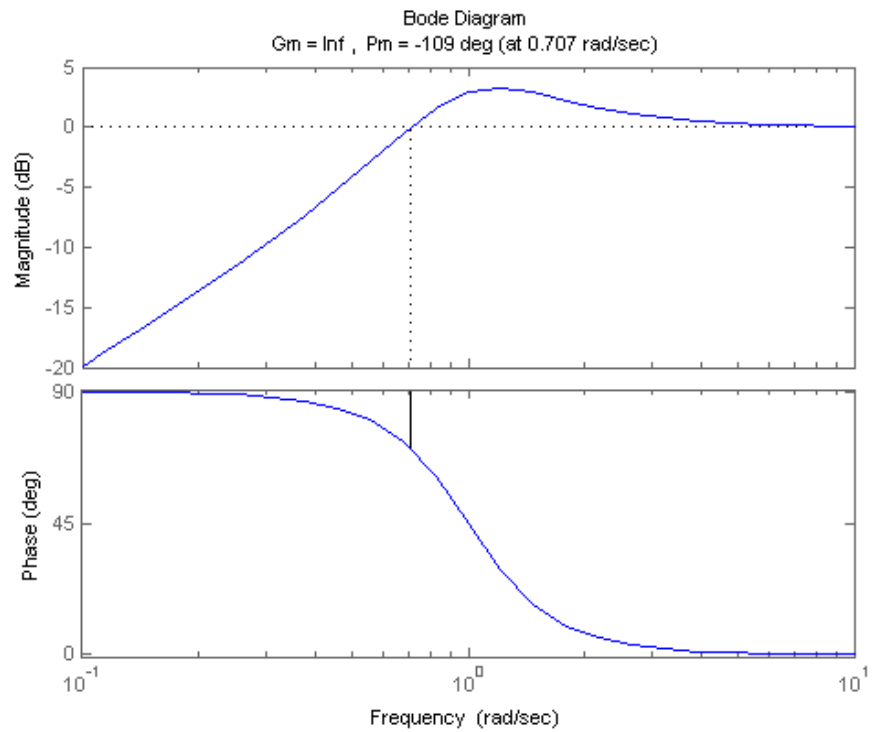
- Let **C(s) = 1**, so that kL(s) = G(s)



T(s)


Bode Diagram
Gm = Inf dB (at Inf rad/sec) , Pm = 90 deg (at 1 rad/sec)
Magnitude (dB)
Phase (deg)
Frequency (rad/sec)

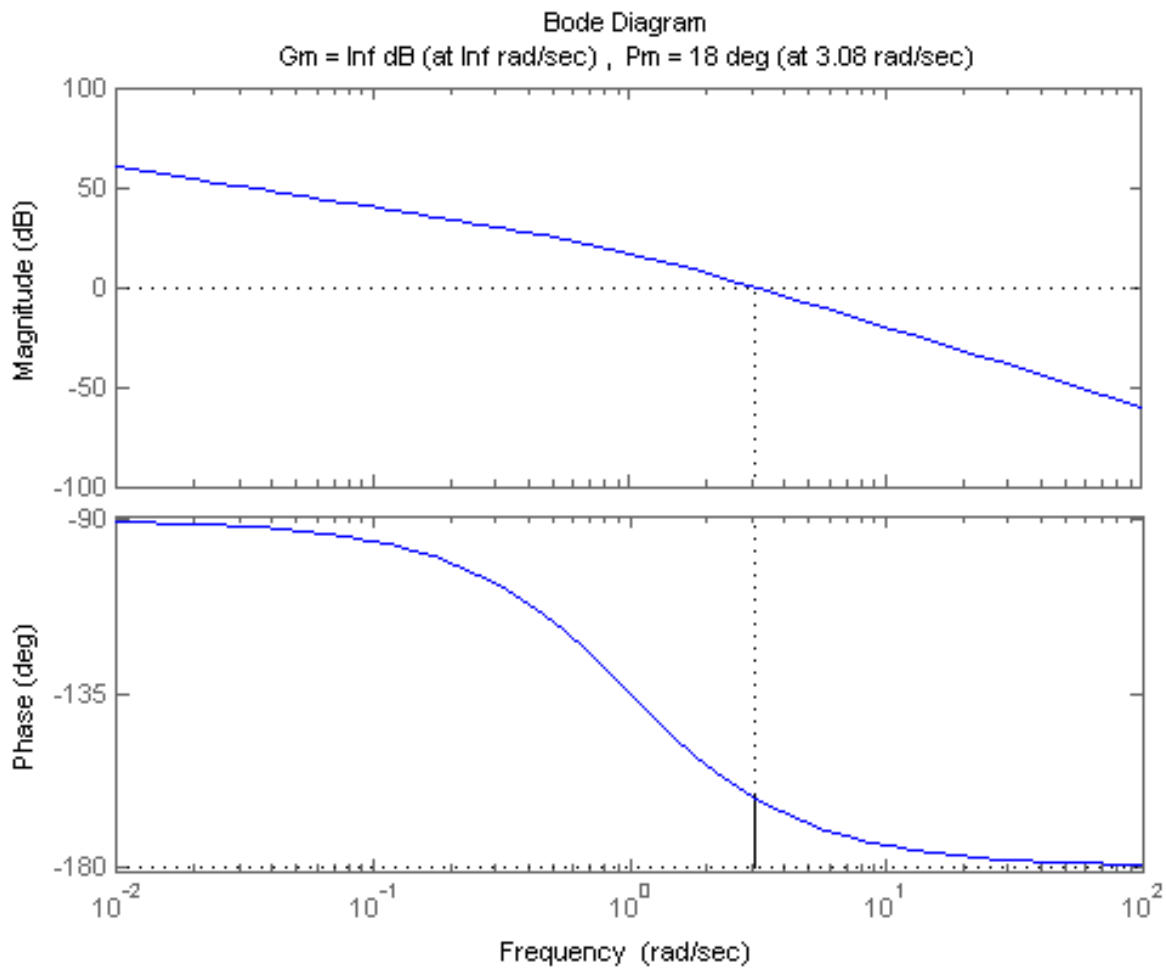
S(s)


Bode Diagram
Gm = Inf , Pm = -109 deg (at 0.707 rad/sec)
Magnitude (dB)
Phase (deg)
Frequency (rad/sec)

Example

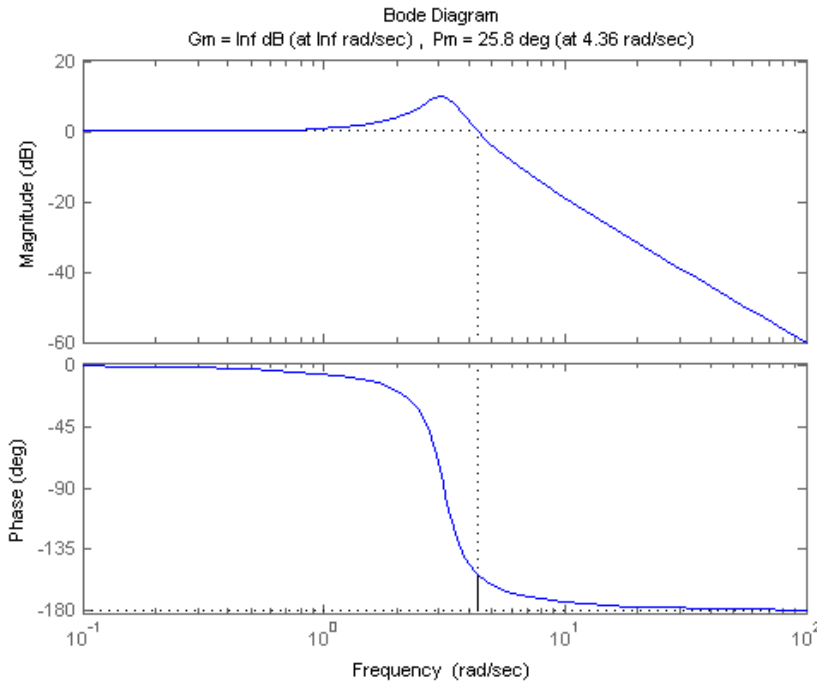
$$G(s) = \frac{1}{s(s+1)}$$

- Let **C(s) = 10**, so that kL(s) = 10.G(s)

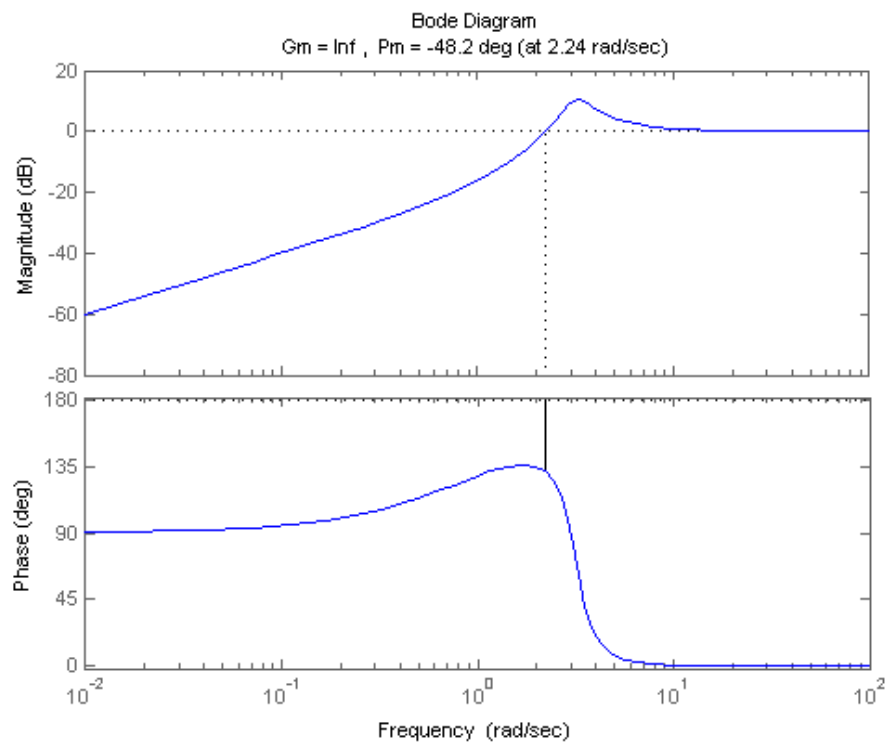


Let **C(s) = 10**, & kL(s) = 10.G(s)

T(s)



S(s)



Design specification

- Cross-over frequency should be 3 rad/s but the phase margin should be at least 50°. We need a phase lead compensator with a maximum of 32° phase advance at 3 rad/s.

Design

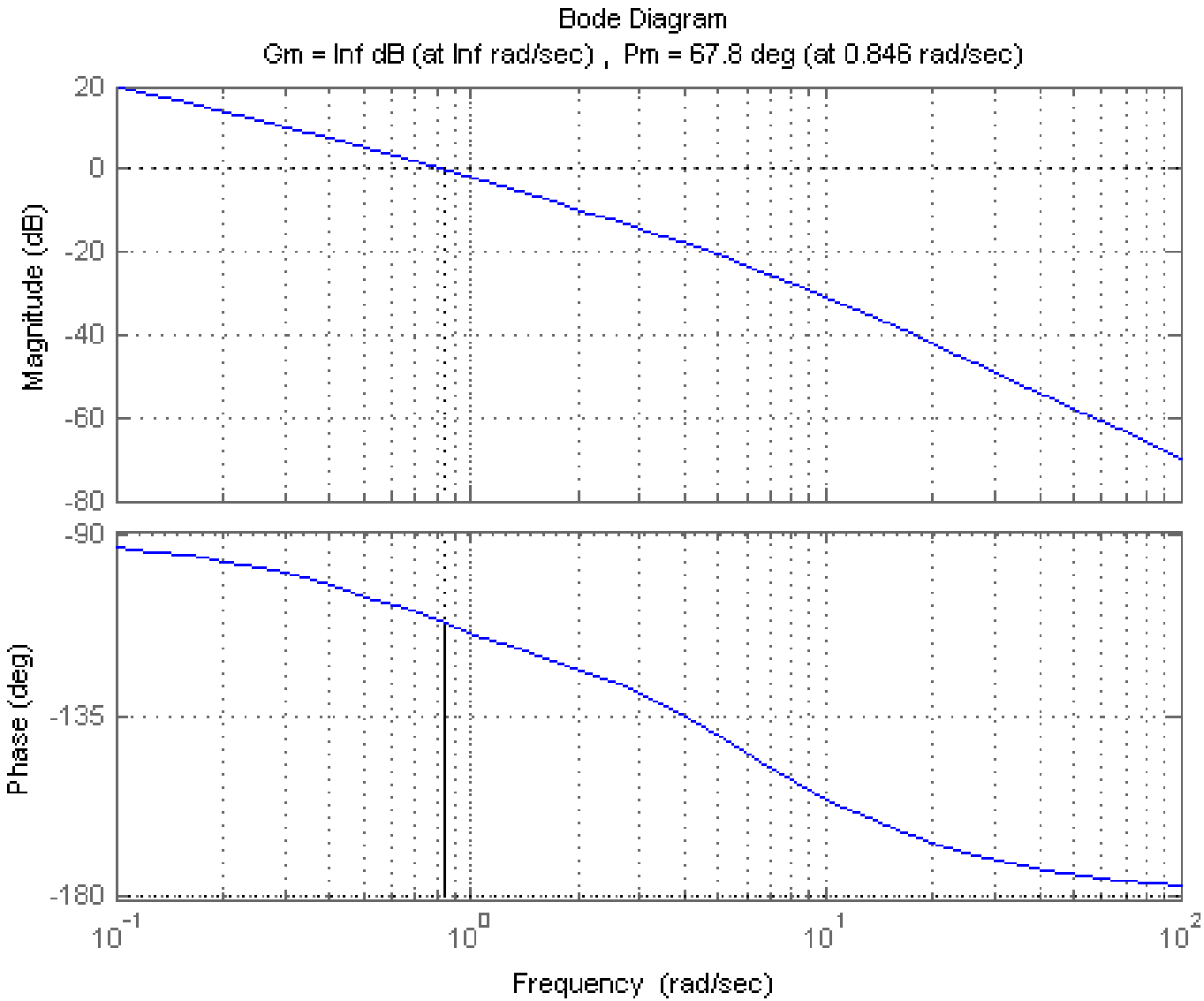
$$\alpha = \frac{1-\sin 32^0}{1+\sin 32^0} \approx 0.31$$

$$\omega_m = 3$$

$$\omega_L = \omega_m \sqrt{\alpha} \approx 1.66$$

$$C(s) = k\frac{\omega_L + s}{\omega_L + \alpha s} \approx k\frac{1.66+s}{1.66+0.31s}$$

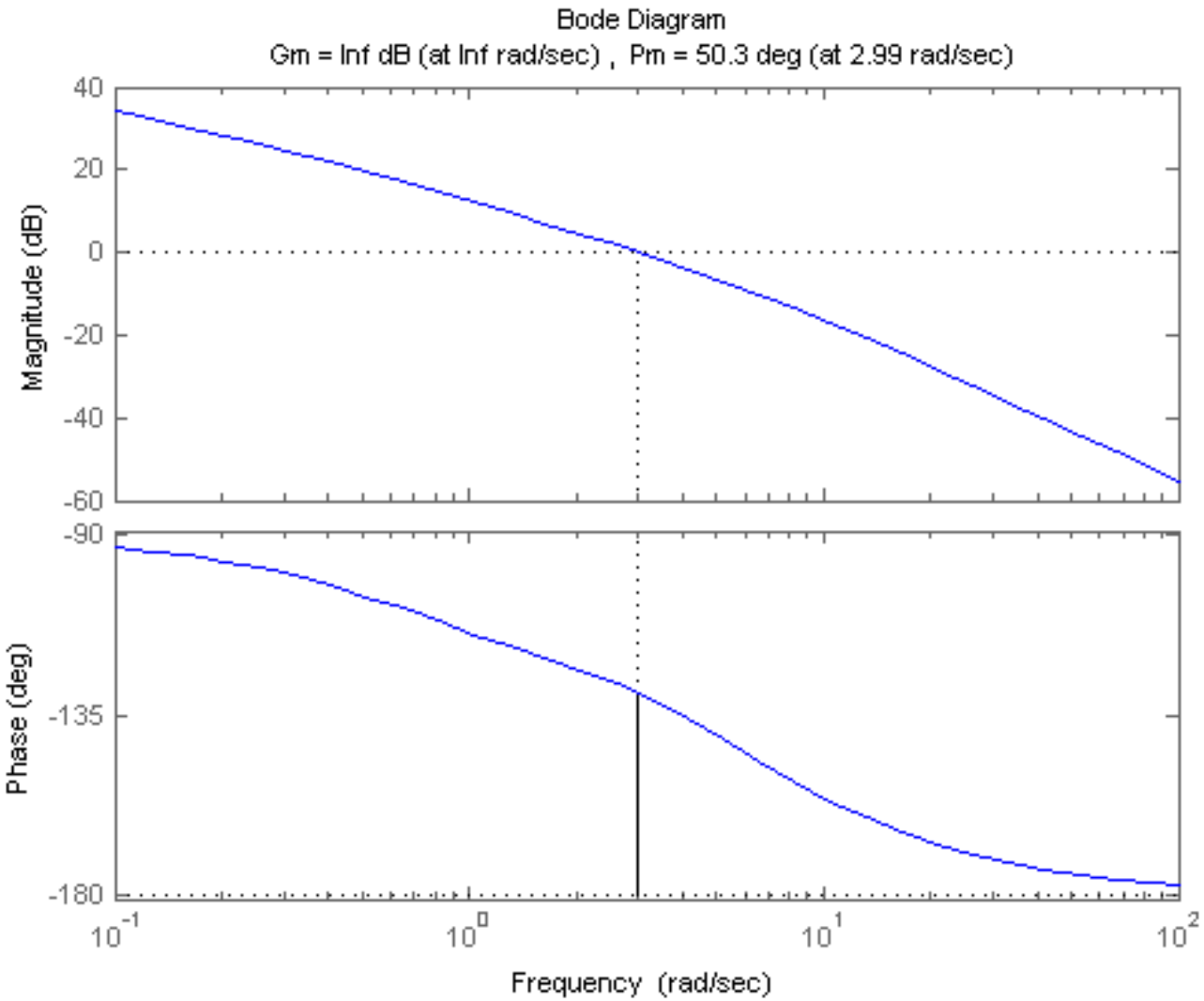
With k =1, C(s).G(s)=kL(s)



Adjusting *k*

- To attain the cross-over frequency $\omega_c = 3$, we need to raise k to 5.25 (gain of G_C with k = 1 is -14dB, at $\omega_c = 3$).
- The phase margin is then 50° as required.

With $$C(s) = 5.25\frac{1.66 + s}{1.66 + 0.31s}$$



Phase lag design

- A phase lag compensator has the same structure as a phase lead compensator but with $\alpha > 1$

$$C(s) = k\frac{1+T_L(s)}{1+\alpha T_L(s)}$$

- It "buys" extra gain at low frequencies, at the "expense" of increased phase. In this sense its role is similar to that of a PI controller.

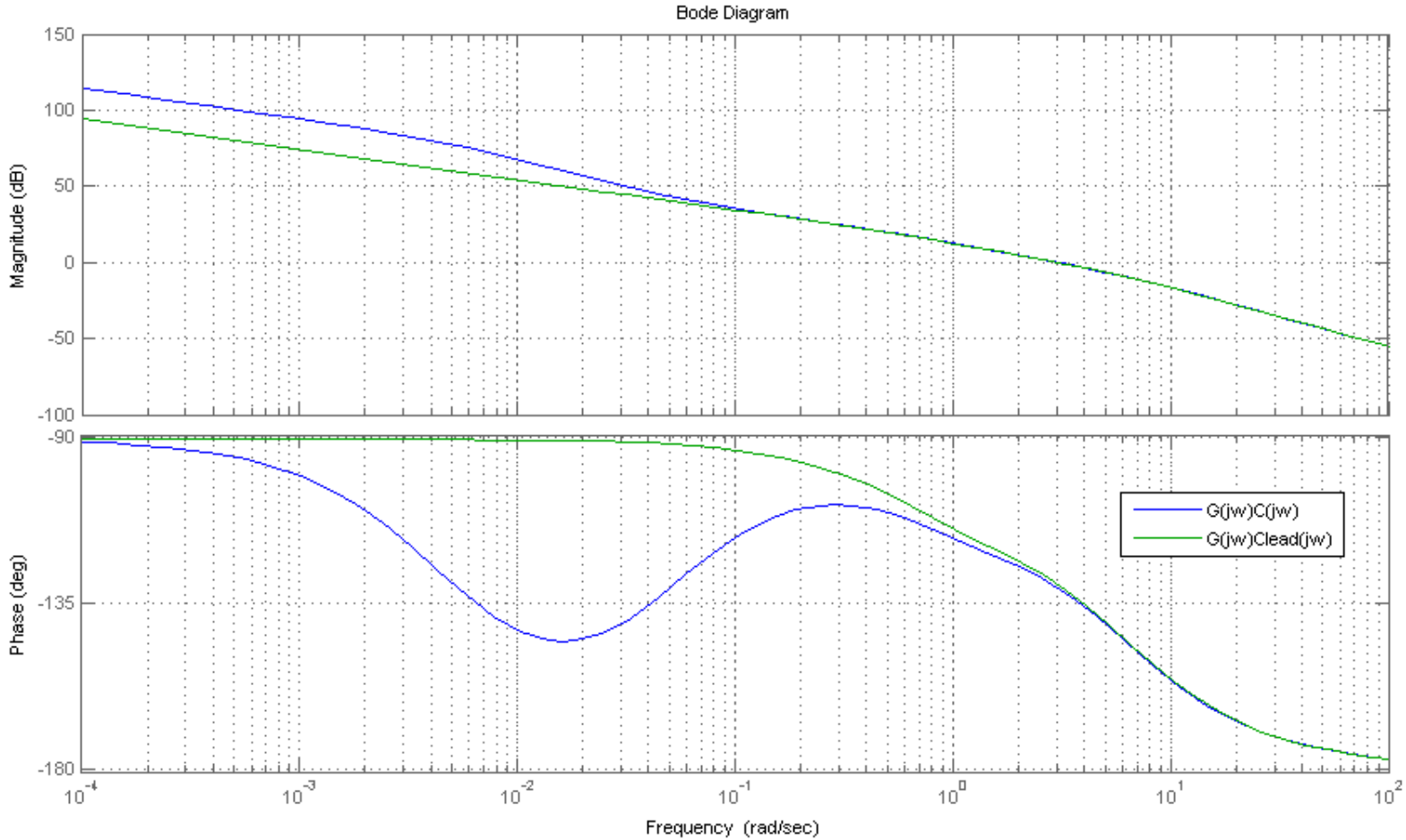
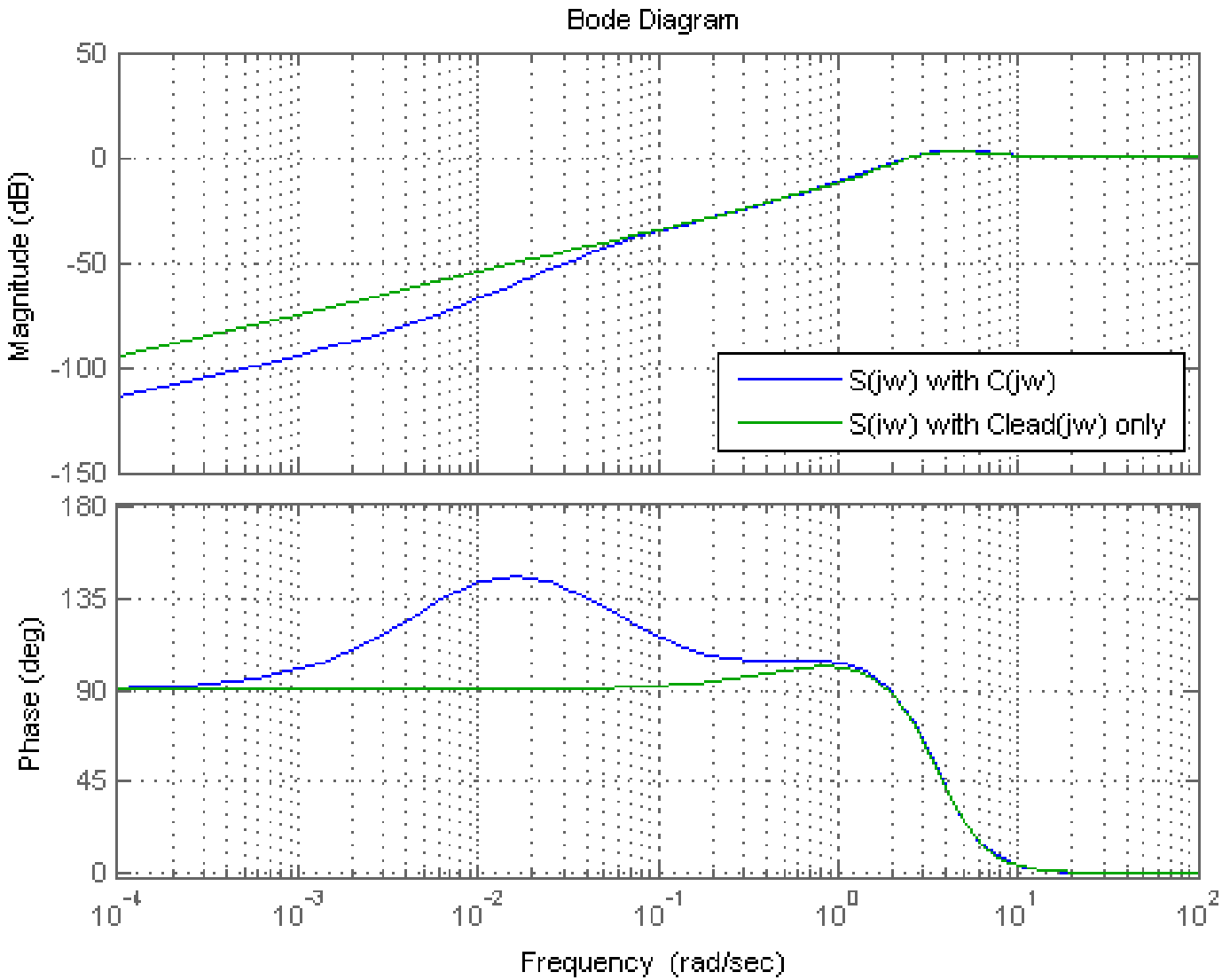
Phase lag controller

$$C(s) = k\,\frac{s+\omega_L}{\alpha s+\omega_L}$$

- We should choose ω_L well below the cross-over frequency, so that the phase margin is not affected.
- Let $G(s) = \dfrac{1}{s(s+1)}$
- Controller $C(s) = C_{lead}(s)C_{lag}(s)$

$$C_{lead}(s) = 5.25\,\frac{1.66+s}{1.66+0.31s}; \quad C_{lag}(s) = \frac{s+0.05}{s+0.005}$$

$G(j\omega)C(j\omega)$ vs. $G(j\omega)C_{lead}(j\omega)$



Sensitivity Function S(jω)